

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412273
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hagi; Gil et al.

System and method for automated simulation of teeth transformation

Abstract

Provided is a system and a method for image synthesis of dental anatomy transformation. In an aspect, there is provided a method comprising: receiving an input segmentation map; transforming the input segmentation map into an input latent vector using a trained encoder; transforming the input latent vector to an output latent vector using a trained transformer machine learning model; transforming the output latent vector to an output segmentation map using a trained decoder; and outputting the output segmentation map.

Inventors:	Hagi; Gil (Toronto, CA), Keszthelyi; Balazs (Budapest, HU)
Applicant:	Tasty Tech Ltd. (Toronto, CA)
Family ID:	81454788
Assignee:	Tasty Tech Ltd. (Toronto, CA)
Appl. No.:	17/519794
Filed:	November 05, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220148188 A1	May. 12, 2022

Related U.S. Application Data

us-provisional-application US 63199038 20201203
us-provisional-application US 63110708 20201106

Publication Classification

Int. Cl.: G06T7/11 (20170101); G06N3/04 (20230101); G06T7/00 (20170101)

U.S. Cl.:

CPC **G06T7/11** (20170101); **G06N3/04** (20130101); **G06T7/0012** (20130101);
G06T2207/20081 (20130101); G06T2207/30036 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10217195	12/2018	Agrawal	N/A	G06T 7/194
12086991	12/2023	Hagi et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110796636	12/2019	CN	N/A
WO2020/181975	12/2019	WO	N/A
WO-2020173912	12/2019	WO	A61C 13/0004

OTHER PUBLICATIONS

https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123600477.pdf Wei et al, Aug. 2020, TANet. cited by examiner

Towards fully automatic tooth arrangement. cited by examiner

Hao et al. Clinically Applicable System for 3D Teeth Segmentation in Intraoral Scans using Deep Learning. Posted: Nov. 10, 2020 [retrieved Mar. 19, 2024], version 1, 22 pages. Retrieved from <<https://www.researchsquare.com/article/rs-103285/v1>> (Year: 2020). cited by applicant

Wang et al. Learning How to Smile: Expression Video Generation With Conditional Adversarial Recurrent Nets. 1st public instance: Jan. 1, 2020 [retrieved Mar. 19, 2024], IEEE Transactions on Multimedia, vol. 22, Issue: 11, Nov. 2020, pp. 2808-2819. Retrieved from <<https://ieeexplore.ieee.org/abstract/document/8948254>> (Year: 2020). cited by applicant

Zanjani et al. Mask-MCNet: Instance Segmentation in 3D Point Cloud of Intra-oral Scans. Oct. 10, 2019 [retrieved Mar. 19, 2024], Medical Image Computing and Computer Assisted Intervention—MICCAI 2019, Lecture Notes in Computer Science: vol. 11768, pp. 128-139. Retrieved from <https://link.springer.com/chapter/10.1007/978-3-030-32254-0_15> (Year: 2019). cited by applicant

Office Action (Non-Final Rejection) dated Mar. 25, 2024 for U.S. Appl. No. 17/539,386 (pp. 1-12). cited by applicant

Yang et al. OrthoPredictor model-guided deep prediction of teeth alignment. Jan. 12, 2020 [retrieved Jun. 15, 2024], ACM Transactions on Graphics, vol. 39, No. 6, Article 1, 16 pages. Retrieved from <[https://scholars.cityu.edu.hk/en/publications/publication\(71e86577-d47e-4512-9bb7-0570f67d15b7\).html](https://scholars.cityu.edu.hk/en/publications/publication(71e86577-d47e-4512-9bb7-0570f67d15b7).html)> (Year: 2020). cited by applicant

Primary Examiner: Zhang; Fan

Attorney, Agent or Firm: Marks & Clerk

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/110,708 filed on Nov. 6, 2020, and U.S. Provisional Patent Application No. 63/199,038 filed on Dec. 3, 2020, which are each hereby incorporated herein by reference.

TECHNICAL FIELD

(1) The following relates generally to imaging manipulation using machine learning techniques; and more specifically to a system and method for image synthesis of dental anatomy transformation.

BACKGROUND

(2) When communicating dental procedures to patients, especially those that relate to cosmetic dentistry, dentists often have to revert to describing or drawing the results of the procedure, or having to provide a picture of an unrelated person to illustrate the results of the procedure. Such approach is generally disadvantageous because the descriptions and drawings are often crude, or the patient has to use their imagination to effectively fill in the blanks to understand how the results of the procedure will look.

SUMMARY

(3) In at least one broad aspect, there is provided a method for automated simulation of teeth transformation comprising: accessing an input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a group of pixels and associated with a respective categorical label; encoding, using a trained encoder model, the input segmentation map to generate the input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector; and decoding, using a trained decoder, the output latent vector to generate an output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed groups of pixels.

(4) In some embodiments, the teeth, expressed by respective transformed group of pixels, have one or more of a transformed position, transformed orientation or transformed shape.

(5) In some embodiments, the method further comprises: initially generating the input segmentation map by analyzing a plurality of pixels of an input image using a trained segmentation model, wherein at least a subset of pixels, of the plurality of pixels, corresponds to one or more exposed teeth in a subject's intraoral region.

(6) In some embodiments, a segment of the plurality of pixels, in the input image, corresponds to a region outside of the subject's intraoral region, and analyzing the plurality of pixels to generate the input segmentation map further comprises assigning the segment of pixels to an irrelevancy mask, and wherein the segment of pixels assigned to the irrelevancy mask are excluded from the encoding, transforming and decoding and are included in the output segmentation map to define the location and shape of the intraoral region.

- (7) In some embodiments, the trained encoder and trained decoder comprise an autoencoder.
- (8) In some embodiments, the method further comprises training an encoder and a decoder to generate the trained encoder and the trained decoder by: receiving a first segmentation map; encoding, using the encoder, the first segmentation map to generate a latent vector; decoding, using the decoder, the latent vector to generate a second segmentation map; determining a loss function associated with a difference between the first and second segmentation maps; and using the loss function to train the autoencoder.
- (9) In some embodiments, each of the trained encoder and trained decoder use a multi-layer perceptron architecture, and wherein the multi-layer-perceptron architecture for the trained encoder can comprise a plurality of expand then squeeze layers each followed by a LeakyRelu activation except for a last layer that is followed by a TanH function to accommodate for a desired type of latent vector.
- (10) In some embodiments, the trained encoder and trained decoder use a convolution neural network (CNN), wherein for the trained encoder, the activation functions comprise LeakyRelu activation except for a last convolution layer that is followed by a TanH function.
- (11) In some embodiments, the trained encoder converts the input segmentation map of dimensions $L \times M \times S$ into the input latent vector of dimensions $1 \times 1 \times P$, wherein “P” is a dimension count of the input latent vector, “L” and “M” are the dimensions of the input image, and S is the number of semantic categories, and wherein the input latent vector defines an input tensor, and at one or more layers of the CNN, a resolution of the input tensor is reduced by a factor of 2×2 while concurrently doubling a number of kernels up to “P” kernels to generate the input latent vector of dimensions $1 \times 1 \times P$, and the trained decoder has an inverse structure to the trained encoder and converts the output latent vector having dimensions of $1 \times 1 \times P$ to the output segmentation map having dimensions of $L \times M \times S$.
- (12) In some embodiments, transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.
- (13) In another broad aspect, there is provided a system for automated simulation of teeth transformation, comprising: a memory unit for storing an input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a group of pixels and associated with a respective categorical label; a processing unit coupled to the memory unit and being operable to perform a method comprising: encoding, using a trained encoder model, the input segmentation map to generate the input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector; and decoding, using a trained decoder, the output latent vector to generate an output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed tooth characteristics.
- (14) In some embodiments, the teeth, expressed by respective transformed group of pixels, have one or more of a transformed position, transformed orientation or transformed shape.
- (15) In some embodiments, the processing unit being further operable to perform the method comprising: initially generating the input segmentation map by analyzing a plurality of pixels of an input image using a trained segmentation model, wherein at least a subset of pixels, of the plurality of pixels, corresponds to one or more exposed teeth in a subject's intraoral region.
- (16) In some embodiments, a segment of the plurality of pixels, in the input image, corresponds to a region outside of the subject's intraoral region, and analyzing the plurality of pixels to generate the input segmentation map further comprises assigning the segment of pixels to an irrelevancy mask, and wherein the segment of pixels assigned to the irrelevancy mask are excluded from the encoding, transforming and decoding and are included in the output segmentation map to define the

location and shape of the intraoral region.

(17) In some embodiments, the trained encoder and trained decoder comprise an autoencoder.

(18) In some embodiments, the processor unit is further configured to perform the method comprising training an encoder and a decoder to generate the trained encoder and the trained decoder by: receiving a first segmentation map; encoding, using the encoder, the first segmentation map to generate a latent vector; decoding, using the decoder, the latent vector to generate a second segmentation map; determining a loss function associated with a difference between the first and second segmentation maps; and using the loss function to train the autoencoder.

(19) In some embodiments, each of the trained encoder and trained decoder use a multi-layer perceptron architecture, and wherein the multi-layer-perceptron architecture for the trained encoder can comprise a plurality of expand then squeeze layers each followed by a LeakyRelu activation except for a last layer that is followed by a TanH function to accommodate for a desired type of latent vector.

(20) In some embodiments, the trained encoder and trained decoder use a convolution neural network (CNN), wherein for the trained encoder, the activation functions comprise LeakyRelu activation except for a last convolution layer that is followed by a TanH function.

(21) In some embodiments, the trained encoder converts the input segmentation map of dimensions $L \times M \times S$ into the input latent vector of dimensions $1 \times 1 \times P$, wherein "P" is a dimension count of the input latent vector, "L" and "M" are the dimensions of the input image, and S is the number of semantic categories, and wherein the input latent vector defines an input tensor, and at one or more layers of the CNN, a resolution of the input tensor is reduced by a factor of 2×2 while concurrently doubling a number of kernels up to "P" kernels to generate the input latent vector of dimensions $1 \times 1 \times P$, and the trained decoder has an inverse structure to the trained encoder and converts the output latent vector having dimensions of $1 \times 1 \times P$ to the output segmentation map having dimensions of $L \times M \times S$.

(22) In some embodiments, transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.

(23) In another broad aspect, there is provided a non-transitory computer-readable medium storing computer program instructions which, when executed by a computer processor, cause the computer processor to perform a method for automated simulation of teeth transformation comprising: accessing an input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a group of pixels and associated with a respective categorical label; encoding, using a trained encoder model, the input segmentation map to generate the input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector; and decoding, using a trained decoder, the output latent vector to generate an output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed groups of pixels.

(24) These and other embodiments are contemplated and described herein. It will be appreciated that the foregoing summary sets out representative aspects to assist skilled readers in understanding the following detailed description.

Description

DESCRIPTION OF THE DRAWINGS

(1) The features of the invention will become more apparent in the following detailed description in

which reference is made to the appended drawings wherein:

- (2) FIG. 1 is a schematic diagram of a system for image synthesis of dental anatomy transformation, in accordance with an embodiment;
- (3) FIG. 2 is a diagram showing an example of teeth layout and numbering;
- (4) FIG. 3 is an example of an input segmentation map, in accordance with the system of FIG. 1;
- (5) FIG. 4 is an example of an output segmentation map, in accordance with the system of FIG. 1;
- (6) FIG. 5 is a diagram of an example architecture for a transformation module, in accordance with the system of FIG. 1; and
- (7) FIG. 6 is a flow chart of a method for image synthesis of dental anatomy transformation, in accordance with an embodiment.

DETAILED DESCRIPTION

(8) Embodiments will now be described with reference to the figures. For simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the Figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein may be practised without these specific details. In other instances, well-known methods, procedures and components have not been described in detail so as not to obscure the embodiments described herein. Also, the description is not to be considered as limiting the scope of the embodiments described herein.

(9) Various terms used throughout the present description may be read and understood as follows, unless the context indicates otherwise: “or” as used throughout is inclusive, as though written “and/or”; singular articles and pronouns as used throughout include their plural forms, and vice versa; similarly, gendered pronouns include their counterpart pronouns so that pronouns should not be understood as limiting anything described herein to use, implementation, performance, etc. by a single gender; “exemplary” should be understood as “illustrative” or “exemplifying” and not necessarily as “preferred” over other embodiments. Further definitions for terms may be set out herein; these may apply to prior and subsequent instances of those terms, as will be understood from a reading of the present description.

(10) Any module, unit, component, server, computer, terminal, engine or device exemplified herein that executes instructions may include or otherwise have access to computer readable media such as storage media, computer storage media, or data storage devices (removable and/or non-removable) such as, for example, magnetic disks, optical disks, or tape. Computer storage media may include volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of computer storage media include RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by an application, module, or both. Any such computer storage media may be part of the device or accessible or connectable thereto. Further, unless the context clearly indicates otherwise, any processor or controller set out herein may be implemented as a singular processor or as a plurality of processors. The plurality of processors may be arrayed or distributed, and any processing function referred to herein may be carried out by one or by a plurality of processors, even though a single processor may be exemplified. Any method, application or module herein described may be implemented using computer readable/executable instructions that may be stored or otherwise held by such computer readable media and executed by the one or more processors.

(11) The following relates generally to imaging manipulation using deep learning techniques; and more specifically to a system and method for image synthesis of dental anatomy transformation.

(12) The present embodiments advantageously allow digital images of a patient's mouth (for example, during a smile) to be taken as an input and a simulated version of the patient's mouth after a cosmetic dental procedure to be provided as output. Thus, allowing the patient to fully visualize the result of the procedure and help the dentist convince the patient of the merits of undertaking such procedure. The present embodiments overcome substantial challenges in the art; for example, problems associated with whitening of the teeth and problems related to structure (layout) of the teeth within the mouth gap. The present embodiments use machine learning techniques to overcome at least these problems; for example, determining teeth layout idealization with the use of semantical transformation.

(13) Referring now to FIG. 1, a system **100** for image synthesis of dental anatomy transformation, in accordance with an embodiment, is shown. In this embodiment, the system **100** is run on a local computing device. In further embodiments, the local computing device can have access to content located on a server over a network, such as the Internet. In further embodiments, the system **100** can be run on any suitable computing device, for example, a server. In some embodiments, the components of the system **100** are stored by and executed on a single computer system. In other embodiments, the components of the system **100** are distributed among two or more computer systems that may be locally or remotely distributed.

(14) FIG. 1 shows various physical and logical components of an embodiment of the system **100**. As shown, the system **100** has a number of physical and logical components, including a processing unit **102** (comprising one or more processors), random access memory ("RAM") **104**, a user interface **106**, a network interface **108**, non-volatile storage **112**, and a local bus **114** enabling processing unit **102** to communicate with the other components. In some cases, at least some of the one or more processors can be graphical processing units. Processing unit **102** executes an operating system, and various modules, as described below in greater detail. RAM **104** provides relatively responsive volatile storage to processing unit **102**. The user interface **106** enables an administrator or user to provide input via an input device, for example a keyboard and mouse. The user interface **106** also outputs information to output devices for the user, such as to a display **110**. The network interface **108** permits communication with other systems, such as other computing devices and servers remotely located from the system **100**, such as for a typical cloud-based access model. Non-volatile storage **112** stores the operating system and programs, including computer-executable instructions for implementing the operating system and modules, as well as any data used by these services. Additional stored data can be stored in a database **116**. During operation of the system **100**, the operating system, the modules, and the related data may be retrieved from the non-volatile storage **112** and placed in RAM **104** to facilitate execution.

(15) In an embodiment, the system **100** further includes a number of functional modules that can be executed on the processing unit **102**; for example, an input module **118**, a transformation module **120**, a segmenter module **124**, and an output module **126**. In some cases, the functions and/or operations of the modules can be combined or executed on other modules.

(16) In some embodiments of the system **100**, representation of semantics for the teeth in an input image can be accomplished with the use of segmentation maps (SM). In an example, the SMs can be $H \times W \times S$ tensors; where H is the height of the input image, W is the width of the input image, and S is the number of semantical categories (for example, $256 \times 256 \times 31$). In some cases, the semantical categories can use a one-hot encoding map; however, any suitable encoding can be used. In an example, a one-hot categorization can be assigned for each tooth position and for each item in the mouth (for example, tongue). In this example, every (x, y) location in the SM corresponds to an S -wide vector that sums up to '1'. More specifically, all elements of the vector are $[0.0, 1.0]$ real numbers where:

$$\sum_{s=0}^{S-1} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} SM(x,y,s) = 1.$$

(17) Referring to the teeth layout and numbering illustrated in FIG. 2, an example semantic class taxonomy for generating the SM can be as follows: {"irrelevancy_mask", 0}, {"tongue", 1},

{“mouth_interior”, 2}, {“gums_upper”, 3}, {“gums_lower”, 4}, {“tooth_11”, 5}, {“tooth_12”, 6}, {“tooth_13”, 7}, {“tooth_14”, 8}, {“tooth_15”, 9}, {“tooth_16”, 10}, {“tooth_21”, 11}, {“tooth_22”, 12}, {“tooth_23”, 13}, {“tooth_24”, 14}, {“tooth_25”, 15}, {“tooth_26”, 16}, {“tooth_31”, 17}, {“tooth_32”, 18}, {“tooth_33”, 19}, {“tooth_34”, 20}, {“tooth_35”, 21}, {“tooth_36”, 22}, {“tooth_41”, 23}, {“tooth_42”, 24}, {“tooth_43”, 25}, {“tooth_44”, 26}, {“tooth_45”, 27}, {“tooth_46”, 28}, {“unknown_object”, 29}, {“implant”, 30}

Where “unknown object” refers to any foreign object in the mouth that the system **100** cannot identify or is not interested in. “Implant” can refer to any exposed dental implant. In the above example, the actual number of classes is 31. In further examples, with the class labels extending all the way to the posterior molars, the number of classes can be 39; where teeth numbers 17, 18, 27, 28, 37, 38, 47, and 48 are included. In this example, the tooth numbering and names are derived from the FDI World Dental Federation notation.

(18) Semantic classes are generally selected based on identification of precise boundaries for types of “objects” in the input image. The system **100** uniquely labels each tooth as a corresponding specific tooth under the FDI notation, rather than, for example, labeling them all as a generic class “tooth”. In most cases, gums and tongue can be labeled because they are used in synthesis, along with the general “mouth_interior” area to generate a plausible mouth. As described herein, for transformation simulation, the pixels of the image defining the interior of the mouth are replaced by the generated image, so the system **100** generates not only teeth but also other aspects of the mouth, such as gums, inner cheeks, and the like. As described herein, after transformation, the system **100** feeds into a synthesizer, so the system **100** learns a “B” label map that includes all possible classes in an ideal smile; including the shape of the gum tissue. In some cases, since exposed implants and foreign objects might appear in the input image, but may be generally undesirable in the simulated output image, these undesirable classes may be excluded from the output image.

(19) In some cases, aliasing in the segmentation map can cause issues and the use of one-hot encoding can help overcome such issues. Instead of having to learn and perform anti-aliasing approaches, which is not ideal, a semantically correct way to interpolate values is required. As an example, anti-aliasing an edge via a smudging technique, comprising taking an average on the order, may produce an incorrect result if the labels are stored as their direct value. For example, smudging of class 1 and class 11 would produce an average of class 6; which is completely incorrect as it produces a value for class 6, which is a different tooth altogether. In contrast, using one-hot encoding, the result of smudging is a vector (for example, [0.0, 0.5, . . . , 0.5, . . . 0.0]), which is a semantically correct representation of the value for a value on the border of two classes.

(20) In a particular case, the segmenter module **124** uses a variant of the U-Net convolutional artificial neural network to build the SM; however, any suitable machine learning paradigm can be used. In this case, branches of the U-Net are determined along an ever decreasing resolution input image pyramid (for example, from top-to-bottom) and an ever increasing resolution pyramid of segmentation attempts (for example, from bottom-to-top). The output segmentation map can comprise the top, full resolution segmentation.

(21) The transformation module **122** takes as input the segmentation map produced by the segmenter module **124** (referred to as Segmentation A) and, via transformation, outputs an output segmentation map (referred to as Segmentation B) representing an idealized teeth layout. FIG. 3 illustrates an example of Segmentation A and FIG. 4 illustrates an example of a corresponding Segmentation B.

(22) The transformation module **122** advantageously uses learnable dimensionality reduction (compression). During development of the system **100**, the present inventors tried using models operating directly on the pixel domain. However, it was determined that the quality and clarity of teeth boundaries were generally not serviceable. As illustrated in FIG. 5, a pair of networks, called an encoder and a decoder, are used by the transformation module **122** to derive and reconstruct a latent vector representation (referred to as a “code”). The encoder is used for derivation (encoding)

while the decoder is responsible for reconstruction of the segmentation map after compression (decoding). In an example, the code comprises an N -wide vector of $[-1.0, 1.0]$ real numbers, where N can be any suitable vector length; however, any suitable latent vector encoding can be used. (23) The decoder can provide certain additional advantages. Since the transformation itself may not be observable by a user, the decoder can be used for verification purposes; i.e., whether the input, or something reasonably close to the input, can be reconstructed after encoding. Additionally, decoders can be used for interpretation of an output of a semantical (layout) transformation (as described herein).

(24) The transformation module **122** can be trained using a set of training images of mouths with which SMs are generated by the segmenter module **124** and labelled by a user. In an example, the transformation module **122** can have a multi-layer-perceptron architecture; however, any suitable machine learning architecture can be used. In an example, the architecture can comprise a number (ex., 5-7) of expand then squeeze layers (ex., expanding to M wide and compressing back to P wide, wherein “ P ” is a dimension count of an input latent vector), each followed by LeakyRelu activations, except for the last layer, which is followed by a TanH function to arrive at a $[-1.0, 1.0]$ vector.

(25) In some cases, the encoder can use a convolutional neural network (CNN) that, at every convolutional layer, reduces the tensor resolution 2×2 (going down from $L \times M$ initial image resolution) which progressively increases kernel depth (up to P); eventually reaching a vector of $1 \times 1 \times P$, which is the size of the latent vector. In this way, there is two-dimensional instance normalization and spectral normalization of the convolutional kernels. As described above, activations are Leaky ReLUs except for after the last convolutional layer where the activation is TanH to accommodate the desired type of latent vector.

(26) In some cases, the decoder can be structurally the inverse of the encoder, going up from the $1 \times 1 \times P$ vector to the $L \times M \times 31$ SM; using the same kind of activations and normalizations. In some cases, a learnable denormalization can be used after the two-dimensional instance normalization to allow the system **100** to not always revert to a unit norm, but rather allow for learning and normalizing into more useful distributions. Otherwise, there may be an issue of trying to find meaningful distributions out of previously normalized values.

(27) As shown in FIG. 5, the encoder and the decoder can make use of a irrelevancy mask. In many cases, the input images are crops of portraits around the mouth, so many of the pixels are outside the mouth gap. The irrelevancy mask, acting as semantical category 0, is then used to define these areas outside the mouth gap. In some cases, the system **100** operates on $(S-1)$ semantical categories, where semantical category 0 is exempt from processing and its path bypasses most of the network as illustrated. In some cases, semantical category 0 can be looked at as an input to provide useful information. In this way, the irrelevancy mask of Segmentation A remains the same even after compression and stays the same after transformation.

(28) The segmenter module **124** can determine which pixels to include in the irrelevancy mask. The segmenter module **124** uses a trained artificial neural network, as described herein, trained with training images labelled with lip boundary, to determine which pixels are outside the mouth gap (lip boundary). In some cases, the segmenter module **124** can operate on an image pyramid (a stack of images that is a product of recursive 2×2 down-sampling). In this way, the segmenter module **124** starts at the bottom image of least resolution and traverses to the top image of highest resolution. The segmenter module **124** initially assumes everything to be part of the irrelevancy mask, and then gradually recognizes portions of the image that define the lip boundary; and eventually fills in the portions within the lip boundary with more refined judgements, e.g., which particular class each region belongs.

(29) In an embodiment, as illustrated in FIG. 5, training of the system **100** can comprise two separate stages (stage 1 and stage 2). In stage 1, the transformation module **122** trains the encoder and the decoder, where the goal of such training is to determine whether the output of the decoder

is as close as possible to the input segmentation map. Additional constraints to offer improved conditioning of the learning problem may also be supplemented here as appropriate. In stage 2, the transformation module **122** trains a transformer network to transform latent vector A into latent vector B. Comparatively, this is generally a faster process that usually has earlier convergence. During stage 2, the results of the first stage remain constant, and the transformer network does not feed back. The transformer network can be trained with a training dataset of images of before and after dental transformation. The ‘before’ training images are converted to Segmentation Map A and then encoded to latent vector A, while the ‘after’ training images are converted to Segmentation Map B and encoded to latent vector B. The transformer learns a transformation of latent vector A to latent vector B. Then the decoder is used to transform it back to a Segmentation Map B from the latent vector. The learning paradigm can use any suitable approach, for example, stochastic gradient descent.

(30) FIG. **6** is a flowchart illustrating an embodiment of a method **600** for image synthesis of dental anatomy transformation. At block **602**, the input module **120** receives an input image; for example, received from the user interface **106**, from the network interface **108**, or from the database **116**. The input image comprising a mouth with teeth exposed.

(31) At block **604**, the segmenter module **124** builds an input segmentation map (Segmentation Map A) using the input image as input; for example, using a trained U-Net convolutional artificial neural network.

(32) At block **606**, the transformation module **122** uses a trained encoder to transform the input segmentation map into an input latent vector (latent vector A). At block **608**, the transformation module **122** uses a trained transformer network to transform the input latent vector to an output latent vector (latent vector B).

(33) At block **610**, the transformation module **122** transforms the output latent vector into an output segmentation map (Segmentation Map B).

(34) At block **612**, the segmenter module **124** generates a simulated image comprising the mouth using the output segmentation map (Segmentation Map B).

(35) At block **614**, the output module **126** outputs the simulated image, the output segmentation map, or both, to the user interface **106**, to the network interface **108**, or to the database **116**.

(36) In some cases, in order to achieve strong regularization, Regularized Auto-Encoders (RAE) can be used for the encoder and the decoder. While Variational Auto-Encoders (VAE) can be used, RAE is preferable because it incorporates elements of strong regularizations (e.g., spectral normalization) on the decoder but not on the encoder. In some cases, similarity between the input of the encoder and the output of the decoder can be determined by a fixed metric (e.g., L1 distance), and in some cases, application of an adversarial discriminator may also be beneficial. The adversarial discriminator can be similar to the encoder, except that it has an output of $1 \times 1 \times 1$ due to being a loss function implementing the adversarial approach.

(37) In some cases, an additional constraint can be applied by enforcing the encoder to have maximal information/entropy via maximum variety of instances of code within batches. This can be used to resist mode-collapse. In this way, the system **100** can enforce the compressed codes to have maximum entropy. Intuitively, if they do not have maximum entropy, that would mean they could be further compressed, which would contradict the nature of compression relying on the minimum amount of entropy needed.

(38) Advantageously, in some cases, the system **100** can use cycle consistency in the context of the encoder and decoder. In these cases, the transformation module **122** can encode the SMs, decode, and then once again encode the decoded data to determine the difference between the originally encoded data and the later encoded data. The difference between the two can be determined using a real distance function. Ideally, this encoded data would be the same as the originally encoded data (if there was lossless compression). Having this additionally constraint can be used to ensure that the small perturbations of compression loss are defined in such way that its impact in the latent

space is minimal. This is also useful for enforcing that similar segmentation maps should have similar codes, which is a very useful property for conditioning. Advantageously, the encoder would otherwise only receive gradients via the decoder during training, but this approach establishes another source of gradients. It is also advantageous for regularization because the imperfections of the decoder ensure small perturbations in the close vicinity of a training sample.

(39) While the present embodiments provide certain approaches with respect to machine learning, other approaches can be used. For example, U-Net designs whereby the encoder and the decoder are implemented together (downwards branch as encoder, upwards branch as decoder). In another example, synthesis oriented GANs (ex., those that are “conditional”) or classical GANs (ex., Deep Convolutional Generative Adversarial Network (DCGAN)) can be used.

(40) Although the foregoing has been described with reference to certain specific embodiments, various modifications thereto will be apparent to those skilled in the art without departing from the spirit and scope of the invention as outlined in the appended claims. The entire disclosures of all references recited above are incorporated herein by reference.

Claims

1. A method for automated simulation of teeth transformation in a subject, the method comprising: generating an input segmentation map by analyzing a plurality of pixels of an input image using a trained segmentation model, wherein at least a subset of pixels, of the plurality of pixels, corresponds to one or more exposed teeth in the subject's intraoral region; accessing the input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a respective subset of pixels, of the plurality of pixels, and associated with a respective categorical label; encoding, using a trained encoder model, the input segmentation map to generate an input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector, wherein the output latent vector is a compressed representation of an output segmentation map; decoding, using a trained decoder, the output latent vector to generate the output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed groups of pixels; and generating an output image using the output segmentation map, wherein transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.
2. The method of claim 1, wherein the one or more teeth, expressed by respective transformed group of pixels, have one or more of a transformed position, transformed orientation or transformed shape.
3. The method of claim 1, wherein, a segment of the plurality of pixels, in the input image, corresponds to a region outside of the subject's intraoral region, and analyzing the plurality of pixels to generate the input segmentation map further comprises assigning the segment of pixels to an irrelevancy mask, and wherein the segment of pixels assigned to the irrelevancy mask are excluded from the encoding, transforming and decoding and are included in the output segmentation map to define the location and shape of the intraoral region.
4. The method of claim 1, wherein the trained encoder and trained decoder comprise an auto-encoder.
5. The method of claim 1, further comprising training an encoder and a decoder to generate the trained encoder and the trained decoder by: receiving a first segmentation map; encoding, using the encoder, the first segmentation map to generate a latent vector; decoding, using the decoder, the latent vector to generate a second segmentation map; determining a loss function associated with a

difference between the first and second segmentation maps; and using the loss function to train an autoencoder.

6. The method of claim 1, wherein each of the trained encoder and trained decoder use a multi-layer perceptron architecture, and wherein the multi-layer-perceptron architecture for the trained encoder can comprise a plurality of expand then squeeze layers each followed by a LeakyRelu activation except for a last layer that is followed by a TanH function to accommodate for a desired type of latent vector.

7. The method of claim 1, wherein the trained encoder and trained decoder use a convolution neural network (CNN), wherein for the trained encoder, the activation functions comprise LeakyRelu activation except for a last convolution layer that is followed by a TanH function.

8. The method of claim 1, wherein, the trained encoder converts the input segmentation map of dimensions $L \times M \times S$ into the input latent vector of dimensions $1 \times 1 \times P$, wherein "P" is a dimension count of the input latent vector, "L" and "M" are the dimensions of the input image, and S is the number of semantic categories, and wherein the input latent vector defines an input tensor, and at one or more layers of the CNN, a resolution of the input tensor is reduced by a factor of 2×2 while concurrently doubling a number of kernels up to "P" kernels to generate the input latent vector of dimensions $1 \times 1 \times P$, and the trained decoder has an inverse structure to the trained encoder and converts the output latent vector having dimensions of $1 \times 1 \times P$ to the output segmentation map having dimensions of $L \times M \times S$.

9. A system for automated simulation of teeth transformation in a subject, the system comprising: a memory unit for storing an input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a group of pixels and associated with a respective categorical label; a computer processor coupled to the memory unit and being operable to perform a method comprising: generating an input segmentation map by analyzing a plurality of pixels of an input image using a trained segmentation model, wherein at least a subset of pixels, of the plurality of pixels, corresponds to one or more exposed teeth in the subject's intraoral region; encoding, using a trained encoder model, the input segmentation map to generate an input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector, wherein the output latent vector is a compressed representation of an output segmentation map; decoding, using a trained decoder, the output latent vector to generate the output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed groups of pixels; and generating an output image using the output segmentation map, wherein transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.

10. The system of claim 9, wherein the one or more teeth, expressed by respective transformed group of pixels, have one or more of a transformed position, transformed orientation or transformed shape.

11. The system of claim 9, wherein, a segment of the plurality of pixels, in the input image, corresponds to a region outside of the subject's intraoral region, and analyzing the plurality of pixels to generate the input segmentation map further comprises assigning the segment of pixels to an irrelevancy mask, and wherein the segment of pixels assigned to the irrelevancy mask are excluded from the encoding, transforming and decoding and are included in the output segmentation map to define the location and shape of the intraoral region.

12. The system of claim 9, wherein the trained encoder and trained decoder comprise an auto-encoder.

13. The system of claim 9, the computer processor being further operable to perform the method comprising training an encoder and a decoder to generate the trained encoder and the trained

decoder by: receiving a first segmentation map; encoding, using the encoder, the first segmentation map to generate a latent vector; decoding, using the decoder, the latent vector to generate a second segmentation map; determining a loss function associated with a difference between the first and second segmentation maps; and using the loss function to train an autoencoder.

14. The system of claim 9, wherein each of the trained encoder and trained decoder use a multi-layer perceptron architecture, and wherein the multi-layer-perceptron architecture for the trained encoder can comprise a plurality of expand then squeeze layers each followed by a LeakyRelu activation except for a last layer that is followed by a TanH function to accommodate for a desired type of latent vector.

15. The system of claim 9, wherein the trained encoder and trained decoder use a convolution neural network (CNN), wherein for the trained encoder, the activation functions comprise LeakyRelu activation except for a last convolution layer that is followed by a TanH function.

16. The system of claim 9, wherein, the trained encoder converts the input segmentation map of dimensions $L \times M \times S$ into the input latent vector of dimensions $1 \times 1 \times P$, wherein “P” is a dimension count of the input latent vector, “L” and “M” are the dimensions of the input image, and S is the number of semantic categories, and wherein the input latent vector defines an input tensor, and at one or more layers of the CNN, a resolution of the input tensor is reduced by a factor of 2×2 while concurrently doubling a number of kernels up to “P” kernels to generate the input latent vector of dimensions $1 \times 1 \times P$, and the trained decoder has an inverse structure to the trained encoder and converts the output latent vector having dimensions of $1 \times 1 \times P$ to the output segmentation map having dimensions of $L \times M \times S$.

17. The system of claim 9, wherein transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.

18. A non-transitory computer-readable medium storing computer program instructions which, when executed by a computer processor, cause the computer processor to perform a method for automated simulation of teeth transformation in a subject, the method comprising: generating an input segmentation map by analyzing a plurality of pixels of an input image using a trained segmentation model, wherein at least a subset of pixels, of the plurality of pixels, corresponds to one or more exposed teeth in the subject's intraoral region; accessing the input segmentation map, the input segmentation map comprising a semantic representation of the subject's intraoral region, wherein in the semantic representation, each tooth in the intraoral region is represented by a respective subset of pixels, of the plurality of pixels, and associated with a respective categorical label; encoding, using a trained encoder model, the input segmentation map to generate an input latent vector, wherein the input latent vector is a compressed representation of the input segmentation map; transforming, using a trained vector transform model, the input latent vector into an output latent vector, wherein the output latent vector is a compressed representation of an output segmentation map; decoding, using a trained decoder, the output latent vector to generate the output segmentation map, the output segmentation map comprising a transformed semantic representation comprising, for one or more teeth, one or more respective transformed groups of pixels; and generating an output image using the output segmentation map, wherein transforming the input latent vector into the output latent vector further comprises the trained vector transform model inserting a representation of one or more simulated teeth in the output latent vector.
